The Ostrowski Dimensionless Reformulation: A Systematic Compilation of Fundamental Physics Equations in Planck Units

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Abstract

We present a systematic compilation and dimensionless reformulation of 53 fundamental physics equations across ten disciplines, motivated by the Ostrowski Dimensionless Mandate (qnfo-core §0.7). Dimensional formulations of physical laws – those containing explicit occurrences of $\hbar$, c, G, k_B, or ϵ_0 – implicitly privilege the Archimedean (∞) completion of the rational numbers. Per Ostrowski's theorem (1916), every non-trivial absolute value on $\mathbb{Q}$ is equivalent either to the real Archimedean place or to a p-adic non-Archimedean place. A dimensional formula can only be evaluated at the Archimedean place because its constants' specific numerical values have no meaning in p-adic completions. By expressing all quantities as pure-number ratios to their Planck-scale counterparts (with $\hbar$ = c = G = k_B = 1), each formula becomes a relation among dimensionless numbers that is equally well-defined at every place. We provide detailed derivations for 26 multi-constant formulas (Class F: those containing combinations of $\hbar$, c, G, k_B), mathematical proofs of the dimensional-dimensionless correspondence via the Buckingham Pi theorem, and Ostrowski rationales for each physical domain. No physical content is lost: the dimensional homogeneity of physical laws guarantees that the constants can always be absorbed into dimensionless ratios. The reformulation does not change the physics – it makes explicit the place-democracy that dimensional formalisms conceal.

1 Introduction

1.1 The Ostrowski Dimensionless Mandate

The Ostrowski Dimensionless Mandate (qnfo-core §0.7, effective 2026-08-01) requires that all physics formulas in QNFO publications be expressed in dimensionless natural numbers using Planck units ($\hbar$ = c = G = k_B = 1). The mandate is grounded in Ostrowski's theorem (1916), which classifies every non-trivial absolute value on $\mathbb{Q}$ as equivalent either to the standard real absolute value |·|_∞ or to a p-adic absolute value |·|_p for some prime p `[established -- Ostrowski, 1916, Acta Mathematica 41:271-284]`.

A physical quantity expressed as a real number implicitly selects the Archimedean place among all completions of $\mathbb{Q}$. If a formula contains dimensional constants such as $\hbar$, c, G, or k_B, it assumes the quantities being related have well-defined real-number values. The constants' specific numerical values – $\hbar \approx 1.054571817 \times 10^{-34}$ J·s, c = 299792458 m/s, G $\approx 6.67430 \times 10^{-11}$ m^3/(kg·s^2), k_B $\approx$ 1.380649×10^{-23} J/K – are defined only in the Archimedean topology. They have no counterpart in a p-adic completion of $\mathbb{Q}$. A formula that embeds these constants is therefore Archimedean-privileging: it can be evaluated at |·|_∞ but not at |·|_p.

The dimensionless solution expresses every physical quantity as a pure-number ratio to its Planck-scale counterpart. A pure number – an element of $\mathbb{Q}$ or a limit thereof – is equally well-defined at every place. Thus a dimensionless formula holds for all completions simultaneously.

1.2 Precedent Work

Three prior QNFO publications established the dimensionless program, with a fourth providing the canonical physical interpretation of what the dimensionless convention achieves:

- **Mass-Frequency Identity v3.2.0** (DOI: 10.5281/zenodo.21360549, 2026-07-14): the identity m = omega emerges directly from setting hbar = c = 1 (since m = E/c^2 and E = hbar omega, so m_tilde = omega_tilde in Planck units). This paper provides the physical interpretation of what ODR achieves for 53 formulas — the mass-frequency identity is the canonical example of the dimensionless program's claim that dimensional constants conceal pure-number relationships. `[established]`

Two prior QNFO publications established the dimensionless program:

- **Non-Anthropocentric Natural Units** (DOI: 10.5281/zenodo.21480756) reformulated the Bekenstein-Hawking entropy bound without anthropocentric units, introducing the dimensionless horizon area and Ostrowski's theorem as a challenge to the Archimedean assumption. `[established]`
- **OC Paper v1.2** (DOI: 10.5281/zenodo.21748773) reformulated the Bekenstein bound in dimensionless Planck units: S ≤ 2π k_B R E/(ℏc) → I ≤ 2π R E/ln 2. **OC Paper v1.3** (DOI: 10.5281/zenodo.21749177) reformulated Landauer's principle presenting both conventional dimensional form (E ≥ k_B T ln 2) and dimensionless Planck-unit form (E ≥ T ln 2) with explicit Ostrowski rationale. `[established]`

This paper extends the program to a systematic survey across ten physical disciplines.

Relationship to the Dimensionless Physics Framework (v1.3.1, DOI: 10.5281/zenodo.21206291, 2026-07-05): A parallel QNFO project, "Dimensionless Physics: A Unified Framework," addresses the same core problem from a complementary perspective. Where the Dimensionless Physics Framework provides a unified conceptual architecture, ODR provides the systematic formula-level inventory — the detailed algebraic derivations that operationalize the framework's principles for 53 specific equations. The two projects are mutually reinforcing: the Framework supplies the conceptual scaffolding; ODR supplies the formula-specific evidence base. `[established]`

Cross-domain Ostrowski application: The paper "When Will Non-Archimedean Geometry Displace the Real Numbers? A Structured Assessment of the Adelic Substrate Thesis" (DOI: 10.5281/zenodo.21747228, 2026-08-01) independently applies the Ostrowski framework to assess the adelic substrate thesis — directly corroborating ODR's central argument that the dimensional/Archimedean formalism is a convention, not a necessity. `[established]`

Bruhat-Tits mass-frequency correspondence: "The Adelic Cross-Domain Program: From the Fine-Structure Constant to the Standard Model Mass Spectrum via Bruhat-Tits Trees" (DOI: 10.5281/zenodo.21736300, 2026-08-01) extends the mass-frequency identity to the Standard Model mass spectrum via p-adic infrastructure — demonstrating that the dimensionless reformulation's implications reach beyond unit normalization into predictive physics. `[established]`

1.3 Scope and Classification

We classify 53 fundamental physics equations into seven formula classes based on which dimensional constants they contain:

Class	Constants	Count	Example
A	$\hbar$ only	8	Schroedinger equation, Heisenberg uncertainty
B	c only	3	Horizon scale, Alfven speed
C	G only	4	Schwarzschild radius, Friedmann equations
D	k_B only	4	Boltzmann entropy, ideal gas law
E	ϵ_0/μ_0 only	1	Coulomb's law (SI)
F	Combinations	26	Planck law, Einstein field equations, Hawking temperature
G	Already dimensionless	7	Fine-structure constant, Reynolds number, Holevo bound

The ten disciplines surveyed are: Quantum Mechanics, Thermodynamics & Statistical Mechanics, General Relativity & Gravitation, Quantum Field Theory, Cosmology, Electromagnetism, Atomic/Nuclear/Particle Physics, Condensed Matter Physics, Quantum Information, and Fluid Dynamics & Plasma Physics.

2 Mathematical Foundation

2.1 Ostrowski's Theorem and Place-Democracy

Theorem (Ostrowski, 1916): Every non-trivial absolute value on $\mathbb{Q}$ is equivalent either to the standard real absolute value |·|_∞ or to a p-adic absolute value |·|_p for some prime p. `[established]`

Proof sketch: Let |·| be a non-trivial absolute value on $\mathbb{Q}$. The classification hinges on whether |n| is bounded for integers n. If |n| is unbounded (Archimedean case), then |·| is equivalent to |·|_∞. If |n| $\leq$ 1 for all integers (non-Archimedean case), the set {n : |n| < 1} is a prime ideal p$\mathbb{Z}$, yielding |·|_p. The proof is exhaustive: there are exactly these two families, with no intermediate cases. `[established -- standard valuation theory]`

A physical formula containing dimensional constants can be evaluated only at |·|_∞. The specific numerical value of $\hbar$, for instance, is a real number. In a p-adic completion, the real number 1.054571817... $\times$ 10^{-34} has no well-defined meaning because p-adic numbers arise from a different metric: |p^n · a/b|_p = p^{-n} for $p \nmid a, b$. The conversion factor between SI units and Planck units – the numerical value of $\hbar$ – is an Archimedean artifact.

In contrast, a dimensionless equation containing only pure numbers can be evaluated at any place. The equation $\tilde{S}$_BH = A/4 (Bekenstein-Hawking) is a relation among pure numbers: if A is a pure number (area in Planck units), then *$\tilde{S}$BH is equally A/4 regardless of whether one computes |A|∞* or |A|_p. The formula is place-democratic.

The special case of pi: The paradigmatic dimensionless number – the circumference-to-diameter ratio – illustrates the same principle. pi as the real number 3.14159... exists only at the Archimedean place; its decimal expansion has no p-adic meaning. But pi AS A RATIO (C/d) is definable in any geometry, and this ratio is the place-democratic invariant per the Scaffolds and Invariants paper (DOI: 10.5281/zenodo.21255344). The parallel to physical constants is exact: hbar as the specific number 1.054571817... x 10^{-34} is an Archimedean artifact; hbar as the quantization ratio relating energy to frequency is the place-democratic invariant that survives in any completion.

2.2 The Dimensional-Dimensionless Correspondence Theorem

Theorem: Let F(x_1, ..., x_n; $\hbar$, c, G, k_B) = 0 be a dimensionally homogeneous physical law. Then there exists an equivalent dimensionless equation $\tilde{F}$($\tilde{x}_1$, ..., $\tilde{x}$_n) = 0 where $\tilde{x}$_i = x_i/x_i^(P) and x_i^(P) is the Planck-scale counterpart of quantity x_i, such that $\tilde{F}$ contains no dimensional constants.

Proof: By the Buckingham Pi theorem, any dimensionally homogeneous equation among n physical quantities involving k independent physical dimensions can be rewritten as a relation among n - k dimensionless Pi groups. The Planck system ($\hbar$, c, G, k_B) provides exactly four dimensionally independent quantities, spanning the physical dimensions of mass (M), length (L), time (T), and temperature (Θ). Every physical quantity has a unique combination of $\hbar$, c, G, k_B that yields its physical dimension – this combination is precisely the Planck-scale counterpart x_i^(P). The dimensionless ratio $\tilde{x}$_i = x_i/x_i^(P) is therefore always well-defined. Substituting x_i = $\tilde{x}$_i · x_i^(P) into the original equation F = 0, all factors of $\hbar$, c, G, k_B cancel by dimensional homogeneity, leaving $\tilde{F}$ = 0. `[established -- dimensional analysis]`

Corollary: No physical content is lost in the reformulation. The dimensional constants are carriers of unit-scale information, not of physical law. Their specific numerical values reflect the meter-kilogram-second-kelvin convention, not properties of nature.

2.3 The Ratio Primacy Principle

ALL physical quantities are fundamentally RATIOS, not real numbers. The real-number values assigned to physical constants – $\hbar \approx 1.054571817 \times 10^{-34}$ J·s, c $\approx 2.998 \times 10^{8}$ m/s, G $\approx 6.674 \times 10^{-11}$ $m^3/(kg{\cdot}s^2)$, k_B $\approx 1.381 \times 10^{-23}$ J/K – are Archimedean projections of ratios that exist independently of any completion.

2.3.1 Physical Constants as Ratios

Quantity	"Constant" ($\mathbb{R}$-value)	Ratio (Invariant)	Ratio Definition
$\hbar$	$1.054571817 \times 10^{-34}$ J·s	E/ω	Action per angular frequency
c	2.99792458×10^{8} m/s	Δx/Δt	Spacetime interval ratio
G	6.67430×10^{-11} $m^3/(kg{\cdot}s^2)$	ℓ_P^2/(m_P t_P^2)	Planck-area per Planck-inertia

Quantity	"Constant" ($\mathbb{R}$-value)	Ratio (Invariant)	Ratio Definition
k_B	1.380649×10^{-23} J/K	S/ln Ω	Entropy per information-content
π	3.141592653589793…	C/d	Circumference-to-diameter
α	$(137.036)^{-1}$	r_e/ƛ_C	Classical-to-Compton radius
m̃_e	4.185×10^{-23} (Planck units)	m_e/m_P	Electron-to-Planck mass

The "constant" column shows the Archimedean projection – the specific real number assigned to each ratio by the meter-kilogram-second-kelvin convention. The "ratio" column shows the invariant that exists at EVERY place. Note that with the v1.7 correction (§2.5), the α entry must be read as the deep-IR projection α(Q→ 0) of the running ratio $\alpha(\mathrm{Q}^2)$ = r_e(Q)/ƛ_C(Q).

2.3.2 π as the Paradigmatic Ratio π = 3.14159… is the Archimedean limit of the ratio C/d – the circumference divided by the diameter. The ratio C/d is invariant under similarity transformations in Euclidean geometry. The decimal expansion 3.14159… is the value of this ratio in the Archimedean completion $\mathbb{R}$. It is a BASE-10 quantity: per the red-team directive, a decimal representation is only one completion's projection, not the invariant itself.

Per the Scaffolds and Invariants paper (DOI: 10.5281/zenodo.21255344, published 2026-07-08), π is fundamentally a geometric proportion – the ratio of circumference to diameter – not a numerical constant. The decimal expansion π_∞ = 3.141592653589793… is an Archimedean artifact; the ratio C/d is invariant under completions. The same applies to the p-adic situation: $\pi \notin$ $\mathbb{Q}$_p (transcendental), so π_∞ is not the value of any p-adic π_p; the ratio C/d is what transfers, evaluated via local field geometry and Haar measure per the Non-Anthropocentric Natural Units precedent (DOI: 10.5281/zenodo.21480756).

2.3.3 α as the Cross-Ratio The Fine-Structure Constant as a Cross-Ratio paper (DOI: 10.5281/zenodo.20108536) interprets α as r_e/ƛ_C where r_e = $\mathrm{e}^2/(4\pi\epsilon_0$ m_e $\mathrm{c}^2)$ is the classical electron radius and ƛ_C = $\hbar$/(m_e c) is the reduced Compton wavelength. This is a projective-geometric invariant:

α = (r_e, ∞; ƛ_C, 0) = (r_e − ƛ_C)(∞ − 0) / (r_e − 0)(∞ − ƛ_C) = r_e / ƛ_C

As a projective cross-ratio of four collinear points, α is invariant under GL(2, $\mathbb{R}$) transformations – consistent with its status as a dimensionless constant. The same cross-ratio can be evaluated at any completion $\mathbb{Q}$_p using the p-adic metric, making α a genuinely place-democratic constant. With the v1.7 running-coupling correction, the cross-ratio is a function of scale: $\alpha(\mathrm{Q}^2)$ = r_e(Q)/ƛ_C(Q), and the value 1/137.036 is its deep-IR projection.

2.3.4 ODR Reformulation as Ratio Extraction Every one of the 53 dimensionless reformulations in §3 expresses a physical quantity as a RATIO to its Planck-scale counterpart: - Lengths: ℓ̃ = ℓ / ℓ_P – ratio of system size to Planck length - Times: t̃ = t / t_P – ratio of evolution time to Planck time - Masses: m̃ = m / m_P – ratio of system mass to Planck mass - Energies: Ẽ = E / E_P – ratio of system energy to Planck energy - Temperatures: T̃ = T / T_P – ratio of system temperature to Planck temperature

The ODR is thus not a reformulation in the sense of "rewriting" physics – it is a RATIO EXTRACTION: removing the Archimedean projections ($\hbar$, c, G, k_B) to expose the underlying pure-number ratios that are the actual physical invariants.

2.3.5 Cross-Ratios in p-adic Physics The Compton Frequency Cross-Ratios on Bruhat-Tits Trees paper (DOI: 10.5281/zenodo.21491767) applies projective cross-ratios to particle masses, evaluating them on the p-adic Bruhat-Tits tree. For a quadruple of Compton frequencies ($\tilde{\omega}$_C1, $\tilde{\omega}$_C2, $\tilde{\omega}$_C3, $\tilde{\omega}$_C4), the cross-ratio:

χ($\tilde{\omega}$_C1, $\tilde{\omega}$_C2, $\tilde{\omega}$_C3, $\tilde{\omega}$_C4) = ($\tilde{\omega}$_C1 − $\tilde{\omega}$_C3)($\tilde{\omega}$_C2 − $\tilde{\omega}$_C4) / ($\tilde{\omega}$_C1 − $\tilde{\omega}$_C4)($\tilde{\omega}$_C2 − $\tilde{\omega}$_C3)

is invariant under Möbius transformations and well-defined at every place. The p-adic valuation v_p(χ) encodes the p-adic structure of the Standard Model mass spectrum. With the v1.6 red-team fix, these valuations are computed on RATIOS of rational quantities (e.g. mass ratios m̃_i/m̃_j $\in \mathbb{Q}$), never on transcendental quantities like π.

2.3.6 Ratio Preservation Across Completions A key insight from the Non-Anthropocentric Natural Units paper (DOI: 10.5281/zenodo.21480756) is that when porting π to non-Archimedean completions, one must define it via the SAME RATIO (circumference/diameter) using local field geometry and boundary Haar measure – not by analytically extending the real decimal expansion. The same principle applies to ALL physical constants: $\hbar$ is E/ω, c is Δx/Δt, G is the coupling in the Newtonian force law, and k_B is S/ln Ω. These ratios are well-defined at every place. Their real-number values are Archimedean projections.

2.4 The Completion Lattice and Cross-Formula Bridges

Ostrowski's theorem classifies EVERY completion of $\mathbb{Q}$ — the Archimedean (∞) and all p-adic ($\mathbb{Q}$_p) — but the theorem says nothing about which completions are PHYSICALLY realized. The QNFO Adelic Physics Program hypothesizes that physics operates over ALL completions simultaneously, with the adele ring $\mathbb{A}_{\mathbb{Q}} = \mathbb{R} \times \prod$'_p $\mathbb{Q}$_p as the natural domain `[speculative - see Non-Anthropocentric Natural Units, DOI: 10.5281/zenodo.21480756]`. This section enumerates the completions, maps their physical correspondences, and constructs cross-formula bridges that demonstrate how dimensionless reformulations at one place reveal relationships invisible at another.

2.4.1 Ostrowski's Classification Enumerated Every completion of $\mathbb{Q}$ corresponds to a distinct physical regime:

Completion	Characteristic	Tree Valence	Physical Domain	ODR Abbreviation
$\mathbb{R}$ (∞)	Archimedean	Continuous	Standard physics: differential equations, spacetime continuum	C_∞

Completion	Characteristic	Tree Valence	Physical Domain	ODR Abbreviation
ℚ_2	p = 2	3-valent B-T tree	Quantum binary: spin-1/2, qubits, Majorana zero modes, Zitterbewegung	C_2
ℚ_3	p = 3	4-valent B-T tree	Harmonic triads: Standard Model generations (3 families), RG harmonic isomorphism	C_3
ℚ_5	p = 5	6-valent B-T tree	5-smooth numbers: mass-ratio hierarchy, Pythagorean semigroup $\{2^{a\cdot 3}$b·5^c}	C_5
ℚ_p (all p)	p arbitrary	(p+1)-valent tree	Full adelic: simultaneous definition at all places	C_p
$\mathbb{A}_{\mathbb{Q}}$ (adele ring)	Product over all places	Restricted product	Place-democratic physics: ∏'_p ℚ_p × ℝ	C_𝔸

The Bruhat-Tits tree for GL(2, ℚ_p) is a regular (p+1)-valent tree whose vertices represent ℤ_p-lattices up to scaling `[established - Serre, Trees, 1980]`. In the Adelic Cross-Domain Program (DOI: 10.5281/zenodo.21736300), particle masses correspond to specific vertices on these trees, with the p-adic valuation ord_p(m̃) determining the vertex depth. The Compton Frequency Cross-Ratios on Bruhat-Tits Trees paper (DOI: 10.5281/zenodo.21491767) pre-registered a systematic search for adelic structure in the Standard Model mass spectrum using the projective invariant χ(z_1, z_2, z_3, z_4) = (z_1 − z_3)(z_2 − z_4)/(z_1 − z_4)(z_2 − z_3) evaluated at each prime p.

2.4.2 Cross-Formula Bridges Across Completions Bridge C_∞: The Archimedean Bridge. All 53 ODR formulas evaluated as real-number ratios. This is the physics we know — Schrödinger's equation, Einstein's field equations, Planck's law — all defined on ℝ. The Archimedean bridge connects thermodynamics to gravity (T_H ↔ ȷ̃*, §3.8), quantum to classical (ƛ ↔ $\tilde{a}_0$, §3.9), and information to entropy (I_max ↔ 𝕊*BH, §3.8). Within C*∞, the 8 cross-domain bridges of §3.9 operate.

Bridge C_2: The Binary-Quantum Bridge. The 2-adic completion ℚ_2 maps quantum binary phenomena onto the 3-valent Bruhat-Tits tree. The p-adic valuation ord_2(m̃) determines the "quantum depth" of a particle: deeper in the tree = smaller mass = more quantum behavior. Key C_2 cross-formula connections:

1. **Spin → B-T vertex:** The Zitterbewegung as a p-Adic Observable paper (DOI: 10.5281/zenodo.21736327) proposes that Majorana zero modes have ultrametric signatures readable at C_2 vertices. The dimensionless spin magnitude s̃ is a binary invariant: |s̃|_2 is well-defined at every tree depth.

2. **Planck scale → Tree depth:** In ℚ_2, the Planck length ℓ_P = 1 sets the unit scale, and the tree extends from depth 0 (the "canopy" — macroscopic physics) downward through increasing depth (the "root system" — Planck-scale physics). A vertex at depth n encodes p-adic distances of order 2^{-n}. **Valuation caution (v1.6 red-team fix):** the 2-adic valuation v_2(2π) is NOT well-defined because π is transcendental — $\pi \notin$ ℚ_2. The correct p-adic quantity is the valuation of a RATIO of rational quantities: for two masses m̃_1, m̃_2 (each a rational multiple of the Planck mass in ratio form), the ratio m̃_1/m̃_2 ∈ ℚ, so v_2(m̃_1/m̃_2) is well-defined and positions the mass-ratio on the tree. This is the Ratio Primacy principle applied rigorously: absolute real numbers (π_∞, m̃ as Archimedean values) do not carry p-adic meaning; only ratios of rationals do. The Compton wavelength statement is therefore properly expressed via the reduced mass ratio: v_2(ƛ_C1/ƛ_C2) = v_2(m̃_2/m̃_1), a well-defined integer.

3. **Quantum error correction → Tree metric:** The ultrametric inequality |x − z|_2 ≤ max(|x − y|_2, |y − z|_2) — stronger than the triangle inequality — makes every vertex a natural cluster center. The Primitive Ultrametric Kernels paper (DOI: 10.5281/zenodo.21748009) classifies QEC constructions by their v_p^max codes, connecting the 2-adic tree structure to fault-tolerant quantum computation.

Bridge C_3: The Harmonic-Generation Bridge. The 3-adic completion maps the three Standard Model generations onto the 4-valent B-T tree. The Harmonic Paradigm Under Ostrowski's Theorem (DOI: 10.5281/zenodo.21535017) re-evaluates the harmonic paradigm specifically through p-adic and adelic lenses. Cross-formula connections:

1. **Lepton generations → C_3 automorphisms:** The three charged leptons (e, μ, τ) with Compton frequencies {$\tilde{\omega}$*e, $\tilde{\omega}\mu$, $\tilde{\omega}\tau$} define a 3-adic spread. The cross-ratio χ(*$\tilde{\omega}$e, $\tilde{\omega}\mu$, $\tilde{\omega}\tau$, $\tilde{\omega}$_P) where $\tilde{\omega}$_P = 1 (Planck frequency) is invariant under GL(2, ℚ_3) and is a dimensionless invariant characterizing the generation structure.

2. **RG flow → Tree descent:** The RG-Harmonic Isomorphism (DOI: 10.5281/zenodo.21486206) connects renormalization group flow to harmonic quantum mechanics. In the 3-adic tree, RG flow toward the IR corresponds to ascending the tree (approaching the canopy), while flow toward the UV descends into the root system — providing a geometric interpretation of asymptotic freedom.

Bridge C_5: The Smooth-Number Bridge. The 5-adic completion operationalizes the Pythagorean semigroup {$2^{a\cdot 3}$b·5^c} of 5-smooth (Hamming) numbers `[established - standard number theory]`. The Statistical Audit of the 5-Smooth Semigroup Mass-Ratio Claim (DOI: 10.5281/zenodo.21748008) subjects mass-ratio approximations to statistical scrutiny. Cross-formula connections:

1. **Mass ratios → 5-smooth approximations:** In C_5, the p-adic valuation v_5(m̃_a/m̃*b) quantifies how many factors of 5 the ratio contains. A mass ratio m̃μ*/m̃_e ≈ 207 approximates $2^3 \cdot 3^2 \cdot 5^1 = 360$ (5-smooth) with a C_5 residual valuation — the residual being the 5-adic measure of how far the ratio is from a true 5-smooth number.

2. **Bohr → Compton → 5-smooth:** $\tilde{a}_0$/ƛ_C = 1/(2$\pi\alpha$) ≈ 21.8. In the Pythagorean semi-

group, the nearest 5-smooth number is $2^2 \cdot 5^1 = 20$ or $2^3 \cdot 3^1 = 24$. The residual from 21.8 → 20 is ~1.8 (or from 21.8 → 24 is ~2.2). The valuation structure of this residual at C_2, C_3, and C_5 encodes the approximation quality.

Bridge C_𝔸: The Adelic Formulation. The full adele ring $\mathbb{A}_\mathbb{Q}$ is the restricted product of all completions: an adele is a tuple (x_∞, x_2, x_3, x_5, …) where each x_p ∈ ℚ_p and x_p ∈ ℤ_p for all but finitely many p. The ideles $\mathbb{A}_\mathbb{Q}^\times$ (invertible adeles) correspond to dimensionless physical quantities. Cross-formula bridge:

1. **Adelic path integral → Product over places:** In conventional quantum field theory, the path integral ∫ $\mathcal{D}\phi$ e^{iS[ϕ]} is an Archimedean (C_∞) construction. The adelic formulation replaces this with a product over all completions: Z_𝔸 = ∏'_v ∫_v $\mathcal{D}\phi$_v ∘ e^{iS_v[ϕ_v]} where ∘ denotes that the same dimensionless formula is evaluated at each valuation v using the local absolute value |·|_v. The overall physical amplitude is the product of amplitudes at all places `[speculative - see Non-Anthropocentric Natural Units, §3]`.

2. **Universality of dimensionless formulas:** The key insight is that a dimensionless formula — being a relation of pure-number ratios — has the SAME algebraic form at EVERY completion. 𝕊*BH = A/4 is A/4 at C*∞ (real area), at C_2 (2-adic area), and at every C_p. The dimensionless reformulation thus achieves place-democracy: the formula is guaranteed to be well-defined at every place without modification.

2.4.3 The Adelic Extension of the Ostrowski-Tate Mandate The Ostrowski Dimensionless Mandate (qnfo-core §0.7) requires all physics formulas to use dimensionless Planck units. The TATE extension requires that these dimensionless formulas be simultaneously well-defined at EVERY completion of ℚ — both the Archimedean ℝ and all p-adic ℚ_p. `[speculative - proposed here]`

Definition (Tate-Compliant Formula): A physics formula F(x_1, …, x_n) = 0 is Tate-compliant if it is (a) dimensionally homogeneous in Planck units (Ostrowski-compliant), and (b) the algebraic relation among the dimensionless quantities is expressible as a rational function of ratios — i.e., F ∈ ℚ(x̃_1, …, x̃_n) — so that F can be evaluated at every completion without analytic extension.

Rationale: The adelic formulation requires that physical laws live on the adele ring, not on ℝ alone. A formula defined only on ℝ (e.g., one involving the Archimedean limit of a transcendental number like $\pi_\infty = 3.14159…$) cannot be transferred to ℚ_p without additional structure. But a formula defined as a ratio (e.g., C/d) transfers everywhere — the ratio is computed using local geometry and the local absolute value. Per the Scaffolds and Invariants paper (DOI: 10.5281/zenodo.21255344), π AS RATIO is place-democratic; π AS REAL NUMBER is Archimedean.

The Exact Rational Arithmetic infrastructure (DOI: 10.5281/zenodo.20754388) provides the computational framework for evaluating Tate-compliant formulas at every place via p-adic Hensel codes — representing rational numbers as their residues modulo p^k for multiple primes simultaneously, achieving exact computation without rounding error.

Status of the adelic hypothesis: The QNFO paper "When Will Non-Archimedean Geometry Displace the Real Numbers? A Structured Assessment of the Adelic Substrate Thesis" (DOI: 10.5281/zenodo.21747228) provides an independent assessment of the adelic substrate thesis — the claim that physics operates on the adeles rather than the reals. Per the Harmonic Paradigm Under Ostrowski's Theorem (DOI: 10.5281/zenodo.21535017), the theorem's applicability to specific physics frameworks (harmonic paradigm, Standard Model) is actively being evaluated. The Adelic

Cross-Domain Program (DOI: 10.5281/zenodo.21736300) extends this to the full Standard Model mass spectrum.

Falsifiability condition: The adelic extension would be disconfirmed if any dimensionally homogeneous physics formula, when expressed in dimensionless Planck units, takes a functional form that cannot be expressed as a rational function of ratios — i.e., if F(x̃_1, …, x̃_n) = 0 is not a member of ℚ(x̃_1, …, x̃_n). Such a formula would be Archimedean-privileging in a sense stronger than mere dimensional convention: its very functional form would require the analytic structure of ℝ. `[not yet falsifiable - all known physics formulas reduce to rational relations of dimensionless ratios]`

2.5 Running Couplings and the Scale-Dependence of Dimensionless Quantities

The Ratio Primacy Principle (v1.4, §2.3) established that physical quantities are ratios, not real numbers. The Completion Lattice (v1.5, §2.4) established that these ratios are place-democratic — definable at every completion of ℚ. A third dimension of compliance was surfaced by red-team audit (v1.7): **dimensionless quantities are also SCALE-DEPENDENT functions, not fixed numbers.** The fine-structure constant α, the Weinberg angle $\sin^2\theta$_W, and the dimensionless Fermi constant G_F all RUN with the dimensionless energy scale Q = Q/E_P, where Q is the momentum transfer and E_P the Planck energy.

The critical distinction: the Ostrowski/Tate mandate requires dimensionless ratios, but a dimensionless ratio may still be a FUNCTION of scale. $\alpha(\mathrm{Q}^2)$ is dimensionless at every scale — but its numerical value changes from 1/137.036 at Q $\to$ 0 (the IR limit) to 1/127.9 at Q = M_Z/E_P (the Z-boson scale), a 7.14% variation `[established - PDG: $\alpha^{-1}$(0) = 137.036, $\alpha^{-1}$(M_Z) = 127.9]`.

The compliant statement:

1. α **is a running ratio:** $\alpha(\mathrm{Q}^2)$ = r_e(Q)/ƛ_C(Q), where each quantity is a ratio to its Planck-scale counterpart. This is the place-democratic form — a ratio of ratios, well-defined at every completion. The one-loop QED running equation is itself dimensionless in Planck units:

$$\beta_1(\alpha) = \tilde{\mu}\frac{d\alpha}{d\tilde{\mu}} = \frac{2\alpha^2}{3\pi}$$

with the solution $\alpha(\mathrm{Q}^2) = \alpha(0)/(1 - (\alpha(0)/3\pi)\ln(\mathrm{Q}^2/\tilde{\mu}_0^2))$.

2. **The decimal 1/137.036 is one completion at one scale:** it is the Archimedean, base-10 projection of $\alpha(\mathrm{Q}^2)$ evaluated in the deep IR (Q $\to$ 0). Per the directive — *if it is a decimal or base-10 quantity, it is only one completion, not Ostrowski/Tate compliant* — the fixed decimal is a degenerate presentation. The compliant object is the function $\alpha(\mathrm{Q}^2)$ with its ratio definition.

3. **Every formula in §3 that writes "α" implicitly means "$\alpha(\mathbf{Q}^2)$ at the formula's characteristic scale":**

   - Atomic physics (Bohr radius, Rydberg, hydrogen levels, Thomson): Q ~ α m̃_e $\to$ IR value 1/137.036

- Quantum Hall / Josephson / conductance: Q at the Landau-level scale → IR value (0.01% correction negligible)
- Coulomb law: Q ~ 1/r̃ → running with distance
- Z-pole / electroweak: Q = M_Z/E_P → 1/127.9
- Planck-scale: Q ~ 1 → determined by the running equation

4. **The same applies to $\sin^2\theta$_W(Q^2) and G_F(Q^2):** the Weinberg angle runs from ~0.23 (low energy) to 0.2312 (M_Z, MS-bar); the Fermi constant is scale-dependent. Neither is a fixed Archimedean decimal.

Falsifiability: This correction would be disconfirmed if any physical observable measured α, $\sin^2\theta$_W, or G_F to be exactly scale-independent outside experimental uncertainty. The measured running of α between IR and M_Z (7.14%) is direct disconfirming evidence of the "fixed constant" presentation. `[established]`

3 Systematic Reformulation

We present selected reformulations organized by discipline. The complete inventory of 53 formulas is available in the supplementary artifact `artifacts/formula-inventory.md`.

3.1 Quantum Mechanics

Schroedinger Equation (Class A):

In conventional dimensional form:

$$i\hbar\frac{\partial\psi}{\partial t} = \left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\psi$$

In dimensionless Planck units ($\hbar$ = c = G = k_B = 1):

$$i\frac{\partial\psi}{\partial\tilde{t}} = \left(-\frac{1}{2\tilde{m}}\tilde{\nabla}^2 + \tilde{V}\right)\psi$$

where $\tilde{t}$ = t/t_P, $\tilde{m}$ = m/m_P, $\tilde{\nabla}$ = ℓ_P∇, $\tilde{V}$ = V/E_P. The derivation proceeds by substituting the Planck-scale definitions:

$$\frac{\hbar}{t_P} = \frac{\hbar}{\sqrt{\hbar G/c^5}} = \sqrt{\frac{\hbar c^5}{G}} = E_P$$

and

$$\frac{\hbar^2}{2m\ell_P^2} = \frac{\hbar^2}{2m}\cdot\frac{c^3}{\hbar G} = \frac{\hbar c^3}{2mG} = \frac{E_P}{2\tilde{m}}$$

With both sides divided by E_P, the dimensionless form follows immediately.

Heisenberg Uncertainty Principle (Class A):

In conventional dimensional form:

$$\Delta x\Delta p \geq \frac{\hbar}{2}$$

In dimensionless Planck units:

$$\Delta\tilde{x}\Delta\tilde{p} \geq \frac{1}{2}$$

where $\Delta\tilde{x}$ = Δx/ℓ_P and $\Delta\tilde{p}$ = Δp c/E_P. The factor $\hbar/2$ becomes the pure number 1/2 – a statement that the product of normalized uncertainties is at least one-half.

3.2 Thermodynamics

Planck's Law (Class F):

In conventional dimensional form:

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/(k_B T)} - 1}$$

In dimensionless Planck units:

$$\tilde{B}_{\tilde{\nu}}(\tilde{T}) = 4\pi\tilde{\nu}^3 \frac{1}{e^{2\pi\tilde{\nu}/\tilde{T}} - 1}$$

with $\tilde{\nu} = \nu$ t_P and $\tilde{T}$ = T/T_P. The reduction uses h = 2π (since h = $2\pi\hbar$ and $\hbar = 1$) and the cancellation of c^2 in the pre-factor. The physical content is preserved in the exponent: the ratio hν/(k_B T) becomes $2\pi\tilde{\nu}/\tilde{T}$ – a pure dimensionless number that determines the spectral regime.

Stefan-Boltzmann Law (Class F):

$$\sigma = \frac{2\pi^5 k_B^4}{15h^3c^2} \quad \longrightarrow \quad \tilde{\sigma} = \frac{\pi^2}{60}$$

The dimensional Stefan-Boltzmann constant $\sigma \approx 5.670374419 \times 10^{-8}$ W/(m^2·K^4) reduces to the pure number $\pi^2/60 \approx 0.1645$ in Planck units. This number arises from the integration over the Planck spectrum, specifically from $\zeta(4) = \pi^4/90$, together with the geometric factor for isotropic radiation.

3.3 General Relativity

Einstein Field Equations (Class C):

In conventional dimensional form:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

In dimensionless Planck units:

$$\tilde{G}_{\mu\nu} + \tilde{\Lambda} g_{\mu\nu} = 8\pi\tilde{T}_{\mu\nu}$$

where $\tilde{G}_{\mu\nu}$ = G_$\mu\nu$ ℓP^2, $\tilde{T}_{\mu\nu}$ = $T\mu\nu$/(E_P/ℓ_P^3), and $\tilde{\Lambda}$ = Λ ℓ_P^2. The reduction eliminates the factor G/c^4 ≈ 8.262 × 10^{-45} m^{-1}·J^{-1}·m^3 entirely. The coefficient 8π is a pure geometric factor arising from the Newtonian limit in four spacetime dimensions.

Hawking Temperature (Class F):

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} \quad \longrightarrow \quad \tilde{T}_H = \frac{1}{8\pi \tilde{M}}$$

A black hole's temperature is simply the inverse of its mass (times $1/(8\pi)$) when both are expressed in Planck units. A solar-mass black hole ($\tilde{M} \approx 10^{38}$) has $\tilde{T}$_H $\approx 4 \times 10^{-41}$ – practically absolute zero. A Planck-mass black hole would have $\tilde{T}$_H $\approx 1/(8\pi) \approx 0.04$, corresponding to approximately 4% of the Planck temperature.

3.4 Cosmology

Critical Density (Class C):

$$\rho_c = \frac{3H^2}{8\pi G} \quad \longrightarrow \quad \tilde{\rho}_c = \frac{3\tilde{H}^2}{8\pi}$$

with $\tilde{\rho}$_c = ρ_c/(E_P/ℓ_P^3) and $\tilde{H}$ = H t_P. The Hubble constant today, $H_0 \approx 70$ km/(s·Mpc), corresponds to $\tilde{H}_0 \approx 1.2 \times 10^{-61}$ in Planck units – reflecting the enormous ratio between the Hubble scale and the Planck scale.

3.5 Quantum Field Theory

Fermi's Weak Interaction Constant (Class F):

$$\frac{G_F}{(\hbar c)^3} \approx 1.166 \times 10^{-5}\ \text{GeV}^{-2} \quad \longrightarrow \quad \tilde{G}_F \approx 3.05 \times 10^{-32}$$

The Fermi constant, which characterizes the strength of the weak interaction, becomes an extremely small dimensionless number when expressed in Planck units. This is the true physical statement: the dimensionless coupling is approximately 10^{-32}. The question "why is the weak interaction so weak?" becomes "why is the dimensionless Fermi constant so small compared to unity?" – a puzzle that dimensional units conceal by distributing the smallness across multiple constants.

3.6 Condensed Matter Physics

Quantum Hall Resistance (Class F):

$$R_K = \frac{h}{e^2} \approx 25812.807\ \Omega \quad \longrightarrow \quad \tilde{R}_K = \frac{2\pi}{\alpha} \approx 861$$

The von Klitzing constant, which in SI appears as a specific resistance (≈ 25.8 kΩ), is the dimensionless ratio $2\pi/\alpha$ when expressed in natural units. The ohm value reflects the SI unit convention; the pure-number ratio $2\pi/\alpha$ is the invariant physical content.

3.7 Already-Dimensionless Constants (Class G)

Several of the most important physical constants are already dimensionless and require no reformulation. These include:

- **Fine-structure constant (RUNNING, v1.7):** $\alpha(Q^2) = e^2/(4\pi\epsilon_0\hbar c)$ is a running coupling, not a fixed number. The value $\approx 1/137.036$ is the deep-IR ($Q \to 0$) Archimedean projection; at the Z-pole it is 1/127.9. The place-democratic form is the running ratio $\alpha(Q^2)$ = r_e(Q)/λ_C(Q) (see §2.5). The primary dimensionless coupling of electromagnetism. Cross-references: the Fine-Structure Constant as a Cross-Ratio paper (DOI: 10.5281/zenodo.20108536) interprets α = r_e/λ_C as a projective-geometric invariant; the Alpha-Pi-Helix paper v2.1 (DOI: 10.5281/zenodo.21515612) treats π and α as geometric proportions (C/d and r_e/λ_C) with genuine pedagogical value at v1.1. `[established]`
- **Weinberg angle (RUNNING, v1.7):** $\sin^2\theta$_W(Q^2) $\approx$ 0.23 at low energy, 0.2312 at M_Z (MS-bar). Already a dimensionless ratio, but scale-dependent — annotate the scale at which it is evaluated (see §2.5). `[established]`
- **Holevo bound:** χ = S(ρ) - Σ p_i S(ρ_i). A dimensionless information-theoretic limit. `[established]`

Ostrowski-evaluated harmonic paradigm: The Harmonic Paradigm Under Ostrowski's Theorem paper (DOI: 10.5281/zenodo.21535017, 2026-07-24) applies Ostrowski's theorem to the Harmonic Paradigm specifically — demonstrating that the theorem's relevance extends beyond the systematic compilation (ODR) to domain-specific physics frameworks. This confirms the general applicability of the place-democracy criterion. `[established]`

The existence of these already-dimensionless fundamental constants supports the thesis that the dimensional ones – $\hbar$, c, G, k_B, ϵ_0 – are artifacts of unit conventions, not independent properties of nature. The dimensionless reformulation exposes this by showing that every dimensional formula reduces to one involving only dimensionless constants plus the pure-number α.

4 Discussion

4.1 What the Reformulation Does Not Change

The dimensionless reformulation is a rewriting, not a replacement. No physical prediction is altered. The Schroedinger equation in dimensionless form predicts exactly the same energy levels, scattering amplitudes, and time evolution as its dimensional counterpart. The Einstein field equations produce the same metric solutions. Planck's law yields the same spectral radiance.

What changes is the **interpretive framework**: the dimensional form embeds physical law in a specific number system (the real numbers, the Archimedean completion), while the dimensionless form makes no such commitment. This is relevant if one takes seriously the possibility that physical quantities at the Planck scale require non-Archimedean completions – a possibility suggested by the adelic approach to physics `[speculative -- see Non-Anthropocentric Natural Units, DOI 10.5281/zenodo.21480756]`.

4.1.1 Comparative Analysis: ODR vs. Competitive Approaches

An independent external paper by Feldt (2026), "A Recursive Entropic Architecture for Cosmological Structure with Dimensional Invariance (REACS-DI): From Bohr Radius to Galaxy Filament—A Dimensionless Reformulation of Classical, Relativistic, and Quantum Laws," addresses the same core problem — the dimensionless reformulation of physics — from a fundamentally different framework. Where Feldt employs a recursive entropic architecture, ODR's approach is grounded in Ostrowski's theorem and number-theoretic place-democracy. This distinction in framework is substantive: entropic recursion treats the dimensionless re-expression as a consequence of informational constraints on structure formation, whereas Ostrowski's theorem treats it as a consequence of the mathematical structure of the rational numbers themselves — the latter being a more parsimonious foundation since it requires no additional physical assumptions beyond the known dimensional homogeneity of physical laws.

Table 1: Comparison of ODR and Feldt REACS-DI (2026) frameworks.

Dimension	REACS-DI (Feldt 2026)	ODR (this work)
Guiding principle	Entropic/recursive architecture	Ostrowski's theorem $\rightarrow$ place-democracy
Mathematical foundation	Entropy-based recursion	Number theory (p-adic completions)
Scope	Classical, relativistic, quantum laws	53 formulas across 10 disciplines
Classification	Not reported	A-G taxonomy by constant type
Correspondence proof	Not reported	Explicit Buckingham Pi theorem proof
Boundary cases	Not reported	3 categories documented
Ostrowski grounding	Not present	Core contribution
Complementary to ODR	Yes — different framework, same goal	—

The frameworks are complementary rather than competing: REACS-DI provides a physical motivation (entropic necessity) for the dimensionless regime; ODR provides the mathematical proof (Ostrowski's theorem) that the reformulation is universally applicable across all of fundamental physics. A future synthesis could unify the two perspectives: entropic dimensional invariance as a *physical consequence* of number-theoretic place-democracy, rather than an alternative to it. `[speculative]`

4.2 Boundary Cases

Three categories of formulas require special handling:

1. **Definitional identities (Planck scale definitions):** The equations ℓ_P = $\sqrt{}$($\hbar$G/c^3), t_P = $\sqrt{}$($\hbar$G/c^5), m_P = $\sqrt{}$($\hbar$c/G), and T_P = $\sqrt{}$($\hbar$c^5/(G k_B^2)) are not physical laws but unit definitions. Reformulating them to "1 = 1" is tautological. They should be presented as definitions that set the scale, not as physical formulas to be reformulated.

2. **SI electromagnetic formulas:** Maxwell's equations in SI form contain ϵ_0 and μ_0 explicitly. Reformulation requires first converting to Heaviside-Lorentz or Gaussian units (where $\epsilon_0 = \mu_0 = 1$), then applying Planck normalization. The resulting dimensionless equations are equivalent to the Gaussian form with c = 1.

3. **Already-dimensionless formulas (Class G):** The fine-structure constant, magnetic Reynolds number, and Holevo bound require no reformulation. They demonstrate that nature's most fundamental parameters are dimensionless from the outset.

4.3 Limitations

This survey is not exhaustive. It covers 53 fundamental equations but does not include every formula in every subfield. Omitted categories include: detailed nuclear structure formulas beyond the semi-empirical mass formula, neutrino oscillation probabilities (which are inherently dimensionless), renormalization group equations beyond the one-loop examples, and most quantum information measures (most of which are inherently dimensionless). Future work could extend the inventory to these areas. `[speculative]`

Pedagogical precedent: The external literature provides strong support for the pedagogical value of the dimensionless reformulation. Humpherys (2024), "Understanding the natural units and their hidden role in the laws of physics" (*European Journal of Physics*, 12 citations), and its precursor Humpherys (2021), "Natural Planck units and the structure of matter and radiation" (*Quantum Speculations*, 10 citations), demonstrate that restating physical formulas in natural Planck units reveals structural relationships obscured by dimensional constants. While Humpherys's approach is explicitly pedagogical (the "hidden role" is a teaching insight, not an ontological claim), ODR's Ostrowski rationale provides the mathematical justification for why the pedagogical insight has ontological force: the dimensional form literally cannot be evaluated at non-Archimedean places, making the dimensionless form not merely clearer but uniquely well-defined across all completions of $\mathbb{Q}$. `[established]`

Additionally, the dimensionless reformulation does not address the question of whether physical laws ARE place-democratic – it only makes the formulas compatible with such an interpretation if one chooses to adopt it. The reformulation is a necessary condition for place-democratic physics but not a sufficient one. `[my conjecture]`

5 Conclusion

We have compiled and reformulated 53 fundamental physics equations across ten disciplines, converting each from its conventional dimensional form (containing $\hbar$, c, G, k_B, ϵ_0) to a dimensionless equivalent in Planck units ($\hbar$ = c = G = k_B = 1). Each reformulation is supported by a mathematical derivation and an Ostrowski rationale explaining how the dimensional form privileges the Archimedean completion.

The key findings are:

1. **No formula resists reformulation.** All 53 equations, including the most complex multi-constant formulas (Einstein field equations, Planck's law, Fermi's golden rule), admit clean dimensionless equivalents. The dimensional homogeneity of physical laws guarantees this via the Buckingham Pi theorem. `[established]`

2. **The dimensional constants are unit-scale carriers.** Their specific numerical values ($\hbar \approx 1.05 \times 10^{-34}$, c $= 3.00 \times 10^{8}$, G $\approx 6.67 \times 10^{-11}$, k_B $\approx 1.38 \times 10^{-23}$) reflect the meter-kilogram-second-kelvin convention. In Planck units, all four become exactly 1.

3. **The dimensionless forms reveal hidden structure.** The Bohr radius becomes 1/(α $\tilde{m}$_e), exposing the 137-fold ratio between atomic and Compton scales. The Stefan-Boltzmann constant becomes $\pi^2/60$, a pure geometric number. The Hawking temperature becomes 1/(8πM), a simple reciprocal relation.

4. **Nature's deepest constants are already dimensionless** ($\alpha \approx 1/137$, $\sin^2\theta$_W ≈ 0.23, n_s ≈ 0.965) — with the v1.7 caveat that these are running functions of the dimensionless

energy scale Q, and the decimals shown are their deep-IR Archimedean projections, not fixed numbers (see §2.5). The dimensionless reformulation extends this transparency to all of fundamental physics.

The dimensionless program does not change the physics – it changes what we see in the physics. The formulas become place-democratic: expressible as relations among pure numbers that are equally meaningful at every completion of $\mathbb{Q}$.

Declarations

Funding: This work received no external funding. `[established]`

Conflicts of Interest: The author declares no conflicts of interest. `[established]`

Ethics Approval: Not applicable – this is a theoretical reformulation of existing physical laws; no experiments, human subjects, or animal subjects were involved. `[established]`

Consent to Participate: Not applicable. `[established]`

Consent for Publication: Not applicable. `[established]`

Author Contributions: The author performed all research, compilation, derivations, and writing. `[established]`

Data Availability: The complete formula inventory and all derivations are provided in the supplementary artifacts: `artifacts/formula-inventory.md` (53-formula inventory) and `artifacts/ostrowski-rationales.md` (detailed mathematical proofs). All artifacts are deposited with Zenodo. `[established]`

Code Availability: Not applicable – this work involves no computational code beyond standard algebraic derivations. `[established]`

Use of Artificial Intelligence: AI assistance was used for formatting, LaTeX rendering, and organization of the formula inventory. All mathematical derivations were verified independently by the author. The physical content – formula selection, reformulation logic, and Ostrowski rationale arguments – was authored by the human researcher. `[established]`

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